

MA5020: Computational Methods for Fluid Flow

Lecture 1: The Third Pillar and the Euler Equations

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1 The Philosophy: The Third Pillar of Science

In the history of science and engineering, we have traditionally relied on two pillars to understand the universe:

1. **Theory:** Analytical solutions, simplified models (e.g., potential flow).
2. **Experiment:** Wind tunnels, scale models, physical measurements.

Today, we welcome the **Third Pillar: Computation (CFD)**. CFD allows us to solve fluid problems that are too complex for theory and too dangerous or expensive for experiments—such as the re-entry of a spacecraft into the atmosphere or the flow of blood through a beating human heart.

2 The CFD Roadmap: From Physics to Bits

In this course, we don't just "run code." We follow a rigorous pipeline:

Physics → Mathematical Model (PDEs) → Numerical Scheme → Code → Results

Today's Goal: We are focusing on Step 2. We will translate the physical laws of conservation into the mathematical language of the **Euler Equations**.

3 The Protagonists: The Three Conservation Laws

Before the math, let's understand the "common sense" of fluids. Every fluid flow must obey:

- **Mass Conservation:** Matter cannot be created or destroyed.
- **Momentum Conservation:** A fluid parcel changes its motion only if a force (like pressure) acts on it.
- **Energy Conservation:** Energy is transformed (between kinetic and internal) but its total amount is fixed.

4 Kinematic Foundations: How we observe

To write these laws mathematically, we use the **Eulerian viewpoint** (fixing our eyes on a point in space). To bridge this with the moving fluid, we define the **Material Derivative**:

$$\frac{D\phi}{Dt} = \frac{\partial\phi}{\partial t} + (\mathbf{v} \cdot \nabla)\phi \quad (1)$$

And the **Reynolds Transport Theorem (RTT)**, which translates laws from a fixed mass (*System*) to a fixed region of space (*Control Volume, CV*):

$$\left. \frac{dB}{dt} \right|_{sys} = \frac{\partial}{\partial t} \iiint_{CV} \rho\beta dV + \iint_{CS} \rho\beta(\mathbf{v} \cdot \mathbf{n}) dA \quad (2)$$

5 Step 1: Deriving Conservation of Mass

1. **Physical Law:** The mass of a fluid system is constant: $\frac{dm}{dt} = 0$.
2. **Apply RTT:** Set $\beta = 1$ (mass per unit mass):

$$\frac{\partial}{\partial t} \iiint_{CV} \rho dV + \iint_{CS} \rho(\mathbf{v} \cdot \mathbf{n}) dA = 0 \quad (3)$$

3. **Divergence Theorem:** Convert the surface flow into a volume integral:

$$\iiint_{CV} \left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) \right] dV = 0 \quad (4)$$

Since this holds for any volume, we get our first PDE:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \quad (5)$$

6 Step 2: Deriving Conservation of Momentum

1. **Physical Law:** $\mathbf{F} = \frac{d\mathbf{P}}{dt}$. For our model, we assume the only force is **Pressure** p :

$$\mathbf{F} = - \iint_{CS} p \mathbf{n} dA = - \iiint_{CV} \nabla p dV \quad (6)$$

2. **Apply RTT:** Set $\beta = \mathbf{v}$ (momentum per unit mass):

$$\frac{\partial}{\partial t} \iiint_{CV} \rho \mathbf{v} dV + \iint_{CS} (\rho \mathbf{v})(\mathbf{v} \cdot \mathbf{n}) dA = - \iiint_{CV} \nabla p dV \quad (7)$$

3. **Result:** Moving all terms to the left under a single integral:

$$\frac{\partial(\rho \mathbf{v})}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} + p \mathbf{I}) = 0 \quad (8)$$

7 Step 3: Deriving Conservation of Energy

1. **Physical Law:** Total Energy E_t changes only due to **Work** done by pressure:

$$\frac{\partial}{\partial t} \iiint_{CV} \rho E_t dV + \iint_{CS} \rho E_t (\mathbf{v} \cdot \mathbf{n}) dA = - \iint_{CS} p (\mathbf{v} \cdot \mathbf{n}) dA \quad (9)$$

2. **Result:** Combining terms:

$$\frac{\partial(\rho E_t)}{\partial t} + \nabla \cdot [\mathbf{v}(\rho E_t + p)] = 0 \quad (10)$$

8 Defining the Euler Equations

When we group these three conservation laws together and assume the fluid has **no viscosity** and **no heat conduction**, we arrive at the **Euler Equations**.

In 1D, they are most elegantly expressed as a unified vector system:

$$\frac{\partial \mathbf{q}}{\partial t} + \frac{\partial \mathbf{f}(\mathbf{q})}{\partial x} = 0 \quad (11)$$

where the **State Vector** $\mathbf{q}$ and **Flux Vector** $\mathbf{f}(\mathbf{q})$ are:

$$\mathbf{q} = \begin{bmatrix} \rho \\ \rho u \\ \rho E_t \end{bmatrix}, \quad \mathbf{f}(\mathbf{q}) = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ u(\rho E_t + p) \end{bmatrix} \quad (12)$$

9 Physical Insights: The Nature of the System

To solve these, we use an **Equation of State** (Ideal Gas Law) to link pressure to energy:

$$p = (\gamma - 1)\rho \left(E_t - \frac{1}{2}u^2 \right) \quad (13)$$

Why Euler? Information and Shocks

Because we ignore viscosity, these equations allow for **shocks**. Information in these equations travels at speeds u and $u \pm a$. This "wave-like" nature is why the Euler system is called a **Hyperbolic** system—the central theme of this course.

10 Summary

We have built the mathematical model for an inviscid fluid. This set of PDEs is the "DNA" of the flows we will simulate.

→ **Next Time:** We will classify these PDEs and discover why solving a "Hyperbolic" equation is fundamentally different from solving an "Elliptic" one.